

D.A.V. LOHAGHAT
CLASS TEST - I (2025-26)
SUBJECT - SCIENCE

MM: 30

CLASS-X

Time: 1:00 hours

BIOLOGY

- Q 1.** Which of the following nutrients are considered as energy sources? [1]
- A. Proteins and mineral salts C. Carbohydrates and fats
B. Water and roughage D. Vitamins and minerals
- Q 2.** Where does food digestion occur in Amoeba? [1]
- A. Mitochondria B. Chloroplast C. Pseudopodia D. Food vacuole
- Q 3.** What specialised structures increase the surface area for nutrient absorption in the human small intestine? [1]
- A. Cilia B. Villi C. Plicae circulares D. Rugae
- Q 4.** Which enzyme is essential for activating enzyme pepsin in the stomach [1]
- Q 5.** List three events that occur during the process of photosynthesis in plants. [2]
- Q 6.** What governs the opening and closing of stomata? [2]
- Q 7.** Differentiate between aerobic and anaerobic respiration. [2]

CHEMISTRY

- Q 8.** Write the chemical formula of the following - [2]
- i. Calcium hydroxide iii. Sulphurous acid
ii. Calcium phosphate iv. Potassium Nitrate
- Q 9.** Explain the following terms- [4]
- i. Valency iii. Isobars
ii. Isotopes iv. Law of conservation of mass
- Q 10.** Balance the following chemical equations- [4]
- i. $\square \text{Al} + \square \text{HNO}_3 \longrightarrow \square \text{Al}(\text{NO}_3)_3 + \square \text{H}_2$
ii. $\square \text{MgCl}_2 + \square \text{Al}(\text{OH})_3 \longrightarrow \square \text{AlCl}_3 + \square \text{Mg}(\text{OH})_2$
iii. $\square \text{ZnFeS}_2 + \square \text{O}_2 \longrightarrow \square \text{Zn} + \square \text{FeO} + \square \text{SO}_2$
iv. $\square \text{P}_4\text{O}_{10} + \square \text{H}_2\text{O} \longrightarrow \square \text{H}_3\text{PO}_4$

PHYSICS

Q 11. A student wants to obtain an erect image of an object using concave mirror of 10 cm focal length. What will be the distance of the object from mirror? [1]

- A. Less than 10 cm C. between 10 and 20 cm
B. 10 cm D. more than 20 cm

Q 12. A student wants to draw a ray diagram to show the formation of image due to a concave mirror when an object is placed in between its focus and centre of curvature. [2]

- i. Write the minimum number of incident rays required.
- ii. State the paths of the incident rays.
- iii. State the paths of the reflected rays.

Q13. (a) Define the following terms in the context of spherical mirrors : [3]

- i. Pole
- iii. Principal axis
- ii. Centre of curvature
- iv. Principal focus

(b) An object is placed at a distance of 60 cm from the pole of a concave mirror. If its real and inverted image is formed on a screen placed at a distance of 60 cm in front of the mirror, what is the focal length of the mirror? Give justification for your answer.

Q 14. Study the data given below showing the focal length of three concave mirrors A, B and C and the respective distances of objects placed in front of the mirrors :

Case	Mirror	Focal Length (cm)	Object Distance (cm)
1	A	20	45
2	B	15	30
3	C	30	20

(a) In which one of the above cases the mirror will form a diminished image of the object? [1]
Justify your answer.

(b) List two properties of the image formed in case 2. [1]

(c) What is the nature and size of the image formed by mirror C? Draw ray diagram to justify your answer. [2]

OR

Name the type of mirror that should be used to obtain the following types of images :

- A magnified and virtual image of an object
- A diminished and virtual image of an object